

Sham Satish Thakare

Independent Computer Science Researcher

Data Engineer / Machine Learning Specialist at Spectra Corporate Service · Pune, India
shamthakare.io · shamthakare3000@gmail.com · [GitHub \(@shamddd\)](https://github.com/shamddd) · [BibTeX](https://arxiv.org/)

Education

APCOER Pune, India

B.Tech. in Artificial Intelligence and Data Science

Cumulative GPA: **8.70 / 10.00**

Core Focus: Machine Learning, Probability & Statistics, Data Structures & Algorithms, Distributed Systems.

01/2020 – 06/2024

Experience

Spectra Corporate Service, India

Data Engineer / Machine Learning Specialist

Engineered high-throughput parallel data processing systems and ML data pipelines using C++, Python, and SQL. Optimized multithreaded cloud workflow automation and telemetry data pipelines. Implemented performance benchmarking across compute clusters.

06/2024 – Present

APCOER Pune, India

Research Intern

Conducted research on machine learning algorithms and statistical data evaluation methods. Provided instructional and teaching support for undergraduate coursework in *Data Structures, Algorithms, and Foundations of Machine Learning*.

06/2023 – 12/2024

Publications

[1] When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in LLMs

Sham Satish Thakare

IEEE Transactions on Artificial Intelligence [[IEEE TAI 2026 \(Submitted\)](#)]

[2] recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

Sham Satish Thakare

IEEE Special Session on Machine Learning on Big Data [[IEEE BigData 2026 \(Submitted\)](#)]

[3] StateShift: State-Matched Evaluation of Error Recovery in Neural Reasoning

Sham Satish Thakare

Elsevier Artificial Intelligence [[Elsevier AIJ 2026 \(Submitted\)](#)]

[4] Amortized Intervention Frontiers for Language-Model Reasoning: When Does Training Beat Search?

Sham Satish Thakare

Transactions on Machine Learning Research [[TMLR Working Paper](#)]

[5] Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints

Sham Satish Thakare

Journal of Machine Learning Research [[JMLR Working Paper](#)]

[6] EnclaveShield: Zero-Knowledge Memory Attestation for Hardware Enclaves

Sham Satish Thakare

IEEE Transactions on Dependable and Secure Computing [[IEEE TDSC Working Paper](#)]

[7] AdaptiveReplica: Dynamic Quorum Adaptation in Distributed Consensus

Sham Satish Thakare

IEEE Transactions on Parallel and Distributed Systems [[IEEE TPDS Working Paper](#)]

[8] TraceMind: Graph-Constrained Causal Reasoning for Root-Cause Localization in Microservice Systems

Sham Satish Thakare

IEEE Transactions on Cloud Computing [[IEEE TCC Working Paper](#)]

Featured Academic Projects and Collaborations

Uncertainty-Weighted Credit Assignment in LM RL

Exposed length confounding ($r = +0.486$) in token predictive entropy, demonstrating uncertainty-weighted GRPO misallocates credit on valid multi-step derivations. Built probing suite `ear_grpo_reasoning`. [IEEE TAI Submitted]

01/2026 – Present

State-Matched Error Recovery Evaluation Protocols

Formulated state-contrast continuation evaluation ($D_{\text{recovery}} = -0.1100$) measuring error recovery under frozen structural covariates across reasoning model rollouts. Built evaluation framework `recovery_eval`. [IEEE BigData Submitted]

02/2026 – Present

Failure-Domain Aware Consensus Quorum Adaptation

Designed C++20 dynamic quorum adaptation for Raft consensus clusters under asymmetric network partitioning. Built open-source library `quorumshift`. [IEEE TPDS Working Paper]

06/2024 – 01/2026

Teaching and Featured Positions

Foundations of Machine Learning & Data Structures, APCOER Pune. *Teaching Assistant*

2023 – 2024

Conducted lab sessions and doubt resolution for undergraduate students. Developed automated grading scripts and algorithmic programming assignments in Python and C++.

Featured Coursework

Mathematics: Advanced Linear Algebra, Probability & Random Processes, Multivariate Calculus, Discrete Mathematics, Numerical Methods.

Machine Learning & Systems: Machine Learning, Deep Learning, Reinforcement Learning, Distributed Systems, Computer Networks, Database Management Systems, Operating Systems.